Dentigo — Analysis and Diagram Design Specification

Version 1.3 | Secretary Workflow and Scope Alignment

Dentigo — Analysis and Diagram Design Specification

Project: Dental Case Quality & Treatment Management System

Purpose: UML Diagrams, ERD, Class Diagram, and Database Design Preparation

Revision Summary

• Added Secretary as a separate system role.

• Patient is a data record only and does not have a system account or login.

• Added Appointment to support appointment management by the Secretary.

• Added Payment to record payments received at the clinic; no online payment gateway is included.

• Clarified that the Doctor alone creates Dental Cases; the Secretary registers patients, manages appointments, accesses system-generated case summaries, and records received payments.

• Administrator manages staff accounts, including Doctor and Secretary accounts.

• Existing dental-case, laboratory, radiology, design-version, and case-history workflows remain in scope.

1. System Actors

1.1 Doctor

• Log In; View Dashboard; Manage Patients; Manage Teeth and Clinical Findings; Manage Treatment Plans; Create Dental Case; Upload Clinical Files; Mark Case as Requiring Radiology; Submit Case to Lab; Review Design; Approve Design; Request Revision; Reject Design; Complete Case; Cancel Case; View Case History; Search Cases; View Notifications.

1.2 Secretary

• Log In; View Secretary Dashboard; Register and Update Patient Information; Create and Manage Appointments; Prepare case progress summaries for the Doctor; Record payments received at the clinic; View permitted patient, appointment, case-summary, and payment information.

• The Secretary cannot make clinical decisions, approve dental designs, or access functions outside the permissions assigned to the role.

1.3 Lab Technician

• Log In; View Lab Dashboard; View Available Cases; View Permitted Case Details; Upload Dental Design; Submit Design for Review; View Design Versions; View Revision Requests; Resubmit Revised Design; View Permitted Case History; View Notifications.

• The Lab Technician cannot access radiology images or reports.

1.4 Radiology User

• Log In; View Radiology Dashboard; View Radiology Queue; View Authorized Case Details; Upload Radiology Images; Upload Radiology Report; View Radiology History; View Notifications.

1.5 Administrator

• Log In; View Administrator Dashboard; Create User Account; Edit User Account; Assign an Existing User Role; Activate or Deactivate Staff Accounts; Monitor Dental Cases; View Case History; Search Cases.

• The Administrator cannot create, cancel, approve, or modify Dental Cases. The Administrator manages staff accounts, including Doctors and Secretaries.

1.6 Patient

• The Patient has no system account and cannot log in. Patient information is registered and managed by authorized clinic staff. The Secretary manages appointment booking and records payments received at the clinic.

1.7 External System Actors

• No external payment gateway, email service, SMS service, or AI service is required in the MVP.

2. Use Case Diagram Specification

Create one system boundary named “Dentigo System”. Place these actors outside the boundary: Doctor, Secretary, Lab Technician, Radiology User, and Administrator. Patient is not an authenticated system actor because the Patient has no account or login.

Doctor Use Cases

Authenticate User; Manage Patient; Manage Clinical Findings; Manage Treatment Plan; Create Dental Case (Doctor only); Validate Case Information; Manage Case Files; Request Radiology; Submit Case to Lab; Review Dental Design; Approve Design; Request Revision; Reject Design; Complete Case; Cancel Case; View Case History; Search Cases; View Notifications.

Secretary Use Cases

Authenticate User; Register Patient; Update Permitted Patient Contact Information; Create Appointment; Manage Appointments; View Automatically Generated Case-Progress Summaries; Record Clinic Payment after receipt; View Payment Records within assigned permissions.

Lab Technician Use Cases

Authenticate User; View Available Cases; View Case Information; Upload Dental

Design; Submit Design for Review; View Design Versions; View Revision Requests;

Resubmit Design; View Permitted Case History; View Notifications.

Radiology User Use Cases

Authenticate User; View Radiology Queue; View Case Information; Upload

Radiology Images; Upload Radiology Report; View Radiology History; View

Notifications.

Administrator Use Cases

Authenticate User; Manage Staff User Accounts; Assign Existing User Role; Activate

or Deactivate Staff Account; Monitor Cases; View Case History; Search Cases.

Suggested Use Case Relationships

• Create Dental Case includes Validate Case Information and is available to the Doctor only.

• Submit Case to Lab includes Validate Case Information.

• Request Revision includes Record Case History.

• Reject Design includes Record Case History.

• Approve Design includes Validate Latest Design Version.

• Cancel Case includes Provide Cancellation Reason.

• Upload Dental Design, Upload Radiology Images, and Upload Radiology Report

each include Validate File.

• Create Appointment includes validating appointment date/time and the

selected patient.

• Record Clinic Payment includes validating the amount, confirming payment was received, and recording the responsible employee.

Use <<include>> only when the included behavior is required as part of the main

use case. Radiology is conditional, not mandatory for every Dental Case.

3. Entity Relationship Diagram (ERD) Preparation

The following are logical data requirements. Final database implementation may

adjust technical details but must preserve the stated business rules.

3.1 User

Attribute Suggested Type Key / Constraint

UserId int PK

FullName nvarchar(150) Required

Email nvarchar(254) Required, Unique

PasswordHash nvarchar(...) Required

Role enum/string Required

IsActive bit Required

CreatedAt datetime Required

3.2 Patient

Attribute Suggested Type Key / Constraint

PatientId int PK

FullName nvarchar(150) Required

PhoneNumber nvarchar(30) Optional

CreatedAt datetime Required

UpdatedAt datetime Optional

3.3 Tooth

Attribute Suggested Type Key / Constraint

ToothId int PK

PatientId int FK → Patient.PatientId

ToothNumber nvarchar(10) Required

Notes nvarchar(max) Optional

3.4 ClinicalFinding

Attribute Suggested Type Key / Constraint

FindingId int PK

PatientId int FK → Patient.PatientId

ToothId int FK → Tooth.ToothId

Nullable FK →

DentalCaseId int

DentalCase.DentalCaseId

Description nvarchar(max) Required

CreatedByUserId int FK → User.UserId

CreatedAt datetime Required

UpdatedAt datetime Optional

3.5 TreatmentPlan

Attribute Suggested Type Key / Constraint

TreatmentPlanId int PK

PatientId int FK → Patient.PatientId

Name nvarchar(150) Required

Description nvarchar(max) Optional

CreatedByUserId int FK → User.UserId

CreatedAt datetime Required

3.6 Procedure

Attribute Suggested Type Key / Constraint

ProcedureId int PK

FK →

TreatmentPlanId int

TreatmentPlan.TreatmentPlanId

Name nvarchar(150) Required

Description nvarchar(max) Optional

Nullable FK →

DentalCaseId int

DentalCase.DentalCaseId

3.7 DentalCase

Attribute Suggested Type Key / Constraint

DentalCaseId int PK

PatientId int FK → Patient.PatientId

ToothId int FK → Tooth.ToothId

TreatmentName nvarchar(150) Required

Description nvarchar(max) Optional

DoctorInstructions nvarchar(max) Optional

Status enum/string Required

RequiresRadiology bit Required

ExpectedDueDate datetime Optional

CreatedByUserId int FK → User.UserId

CreatedAt datetime Required

UpdatedAt datetime Optional

CancellationReason nvarchar(max) Optional

3.8 CaseFile

Attribute Suggested Type Key / Constraint

FileId int PK

PatientId int FK → Patient.PatientId

FK →

DentalCaseId int

DentalCase.DentalCaseId

Nullable FK →

ToothId int

Tooth.ToothId

FileName nvarchar(255) Required

FileType nvarchar(20) Required

FileSize bigint Required

StoragePath nvarchar(...) Required

FileCategory enum/string Required

UploadedByUserId int FK → User.UserId

UploadedAt datetime Required

3.9 DesignVersion

Attribute Suggested Type Key / Constraint

DesignVersionId int PK

FK →

DentalCaseId int

DentalCase.DentalCaseId

FileId int FK → CaseFile.FileId

VersionNumber int Required

SubmittedByUserId int FK → User.UserId

SubmittedAt datetime Required

ReviewStatus enum/string Required

ReviewReason nvarchar(max) Optional

3.10 CaseHistoryEvent

Attribute Suggested Type Key / Constraint

CaseHistoryEventId int PK

DentalCaseId int FK → DentalCase.DentalCaseId

ActorUserId int Nullable FK → User.UserId

EventType nvarchar(100) Required

Description nvarchar(max) Required

PreviousStatus nvarchar(50) Optional

NewStatus nvarchar(50) Optional

RelatedFileId int Nullable FK → CaseFile.FileId

RelatedDesignVersionId int Nullable FK →

DesignVersion.DesignVersionId

Reason nvarchar(max) Optional

CreatedAt datetime Required

3.11 Notification

Attribute Suggested Type Key / Constraint

NotificationId int PK

RecipientUserId int FK → User.UserId

Nullable FK →

DentalCaseId int

DentalCase.DentalCaseId

NotificationType nvarchar(100) Required

Message nvarchar(max) Required

CreatedAt datetime Required

IsRead bit Required

3.12 Appointment

Attribute Suggested Type Key / Constraint

AppointmentId int PK

PatientId int FK → Patient.PatientId

Nullable FK →

DoctorId int User.UserId; role must be

Doctor

FK → User.UserId;

CreatedByUserId int

typically Secretary

AppointmentDate datetime Required

AppointmentStatus

Status Required

enum/string

Notes nvarchar(max) Optional

CreatedAt datetime Required

3.13 Payment

Attribute Suggested Type Key / Constraint

PaymentId int PK

PatientId int FK → Patient.PatientId

Nullable FK →

DentalCaseId int

DentalCase.DentalCaseId

RecordedByUserId int FK → User.UserId

Amount decimal(18,2) Required

PaymentDate datetime Required

PaymentMethod

Method Required

enum/string

PaymentStatus

Status Required

enum/string

Notes nvarchar(max) Optional

CreatedAt datetime Required

4. ERD Relationships and Cardinality

Relationship Cardinality Explanation

Patient — Tooth 1 : Many A patient can have multiple teeth.

Patient — DentalCase 1 : Many A patient can have multiple cases.

Tooth — DentalCase 1 : Many A tooth can have multiple cases over time.

Patient — TreatmentPlan 1 : Many A patient can have multiple treatment plans.

TreatmentPlan — Procedure 1 : Many A plan can contain multiple procedures.

Procedure — DentalCase Optional association A procedure may be linked to a Dental Case.

Patient — ClinicalFinding 1 : Many A patient can have multiple findings.

Tooth — ClinicalFinding 1 : Many A tooth can have multiple findings.

DentalCase — ClinicalFinding 1 : Many, optional on finding A finding may be independent of a case.

DentalCase — CaseFile 1 : Many A case can have multiple files.

DentalCase — DesignVersion 1 : Many A case can have multiple design versions.

CaseFile — DesignVersion 1 : 0..1 recommended Each submitted design version references one CaseFile.

DentalCase — CaseHistoryEvent 1 : Many A case has multiple history events.

User — CaseHistoryEvent 1 : Many, optional actor A user may perform multiple recorded actions.

User — Notification 1 : Many A user may receive

multiple notifications.

DentalCase — 1 : Many, optional link A case may have multiple Notification notifications.

Patient — Appointment 1 : Many A patient can have multiple appointments.

User (Doctor) — 1 : Many Each appointment is assigned Appointment to one Doctor.

User — Appointment 1 : Many A staff user (CreatedByUser) records/creates appointments.

Patient — Payment 1 : Many A patient can have multiple payment records.

DentalCase — Payment 0..1 : Many A payment may optionally be linked to a Dental Case.

User — Payment 1 : Many A staff user records (RecordedByUser) payments.

Integrity rule: DentalCase has both PatientId and ToothId, so the database or backend must ensure that the selected Tooth belongs to the same Patient. An independent foreign key on each column does not automatically guarantee this.

5. Class Diagram Preparation

The Class Diagram represents software classes and their relationships; it is not identical to the ERD.

5.1 Suggested Core Classes

User; Patient; Tooth; ClinicalFinding; TreatmentPlan; Procedure; DentalCase; CaseFile; DesignVersion; CaseHistoryEvent; Notification; Appointment; Payment.

5.2 Suggested Enums

UserRole

Doctor; Secretary; LabTechnician; RadiologyUser; Administrator.

DentalCaseStatus

Draft; Submitted; InProgress; DesignSubmitted; RevisionRequested; Approved; Completed; Cancelled.

FileCategory

Clinical; RadiologyImage; RadiologyReport; DentalDesign; OtherPermittedFile.

DesignReviewStatus

PendingReview; Approved; RevisionRequested; Rejected.

AppointmentStatus

Scheduled; Confirmed; Completed; Cancelled; NoShow.

PaymentMethod

Cash; Card; Other.

PaymentStatus

Paid; Refunded; Voided. A payment record is created only after receipt at the clinic; online payment processing is out of scope.

CancellationReason

CreatedByMistake; WrongPatient; DuplicateCase; TreatmentNoLongerRequired;

Other.

5.3 Important Class Relationships

• Patient contains collections of Tooth, DentalCase, TreatmentPlan, Appointment,

and Payment.

• Tooth contains a collection of DentalCase.

• TreatmentPlan contains a collection of Procedure.

• DentalCase contains collections of CaseFile, DesignVersion, and

CaseHistoryEvent.

• User can create or upload multiple records and receive multiple Notification

records.

• Appointment references one Patient, one Doctor (User), and the staff

User who created it.

• Payment references one Patient, an optional DentalCase, and the User who

recorded it.

• Patient is not a User and has no credentials or login relationship.

6. Sequence Diagram Specifications

6.1 Create and Submit Dental Case

Participants: Doctor (Dental Case creation is Doctor-only); Web UI; Backend API; Database; File Storage; Radiology User only when needed. Main flow: Doctor opens Create Dental Case; selects patient and

tooth; enters treatment details and uploads files; UI sends data to API; backend

validates patient, tooth, and required fields; checks radiology requirement; returns

validation errors if required radiology files are missing; otherwise saves the case;

submits to laboratory when the Quality Gate passes; records Case History; UI

displays result.

6.2 Upload Radiology

Participants: Radiology User; Radiology UI; Backend API; Database; File Storage.

User opens Radiology Dashboard; API returns authorized cases requiring radiology; user selects case and uploads images/report; backend validates permissions, file type, size, and association; files are saved; upload events are recorded; backend checks required files; completed items leave active radiology queue; main Dental Case status remains separate from radiology queue status.

6.3 Submit and Review Dental Design

Participants: Lab Technician; Lab UI; Backend API; Database; File Storage; Doctor. Lab opens an available case; uploads design; backend validates file and permissions; creates a design version; lab submits for review; backend updates case to Design Submitted / Doctor Review, records event, and notifies Doctor; Doctor reviews latest design; approves or requests revision/rejects; backend verifies reviewed version is latest; records decision; revision returns case to laboratory workflow.

6.4 Cancel Dental Case

Participants: Doctor; Web UI; Backend API; Database. Doctor selects Cancel Case; UI requests reason; Doctor selects reason and provides additional text if Other; backend verifies permission and reason; updates status; records event and reason; preserves previous files/history; UI confirms cancellation.

6.5 Create Appointment

Participants: Secretary; Secretary UI; Backend API; Database. Secretary selects a registered Patient, enters appointment date/time and optional Doctor, and submits; backend validates patient, permissions, date/time, and scheduling rules; saves appointment and records creation; UI confirms result.

6.6 Record Clinic Payment

Participants: Secretary or authorized staff; Staff UI; Backend API; Database. Staff selects Patient and optional DentalCase, enters amount, payment date and method after receiving payment at the clinic; backend validates permissions and values; saves payment record with recording user; UI confirms. No online payment gateway is used.

7. Activity Diagram Specifications

7.1 Main Dental Case Activity

Start → Select Patient → Select Tooth → Enter Treatment Details → Attach Required Files → Mark Whether Radiology Is Required → Validate Required Information. If Radiology is required, add case to Radiology Queue → Upload Images and Report → Validate Radiology Files. Then run Quality Gate. If details/files are incomplete, display errors and correct them; otherwise submit

case to Lab → Lab Processes Case → Upload Design → Submit Design for Review → Doctor Reviews Latest Version. Doctor approves or requests revision/rejects; record reason and return to Lab where required. When required work is finished, Doctor marks case Completed and the system records completion. At an allowed stage, Doctor may cancel with a valid reason; preserve files and history.

7.2 Radiology Activity

Start → Open Radiology Queue → Select Authorized Case → Upload Required Images → Upload Report → Validate Files. If files are incomplete, display missing files/errors and upload again. If complete, remove case from active Radiology Queue → Record History → End. Dental Case is not automatically marked Completed.

7.3 Appointment and Payment Activity

Appointment: Secretary selects Patient and Doctor → enters appointment details → validates availability and scheduling rules → saves appointment. Before a visit, the system generates a case-progress summary from existing permitted case data; the Secretary does not manually author clinical content. Payment: patient pays at the clinic → authorized staff records amount/date/method and optional DentalCase → backend validates and saves the record.

8. State Machine Diagram

Use DentalCaseStatus as the main state variable. Suggested transitions: Draft → Submitted; Submitted → InProgress; InProgress → DesignSubmitted; DesignSubmitted → Approved; DesignSubmitted → RevisionRequested; DesignSubmitted → InProgress if rejected and returned to lab; RevisionRequested → InProgress; Approved → Completed when remaining workflow conditions are satisfied; Draft, Submitted, InProgress, DesignSubmitted, RevisionRequested, and potentially Approved → Cancelled only where permitted by the final workflow rules. Backend validates actor, current status, and required conditions before every transition.

Radiology is modeled separately as a queue requirement, such as NotRequired, Required, Submitted. These values do not replace the main Dental Case status.

9. Data and Business Constraints

1. Every Dental Case has exactly one Patient and one Tooth.

2. The selected Tooth must belong to the selected Patient.

3. A Patient may have multiple Dental Cases.

4. A Dental Case may exist without a Treatment Plan.

5. A Treatment Plan may contain multiple Procedures.

6. A Procedure may be linked to a Dental Case.

7. Radiology is conditional, not mandatory for every case.

8. Radiology completion does not complete the Dental Case.

9. Approved does not mean Completed.

10. A new design must not overwrite a previous version.

11. A Doctor review decision must apply to the latest submitted design.

12. Cancellation requires a valid reason.

13. Case History events cannot be edited or deleted through the application.

14. The last active Administrator must be protected.

15. Deactivated users must not log in.

16. Files must pass type and size validation; maximum file size is 25 MB.

17. File access must be validated according to role permissions.

18. A case must leave the active Radiology Queue when all required radiology
files are submitted.

19. The UI cannot be the only enforcement mechanism for business rules.

20. Patient records do not have login credentials or patient accounts.

21. Only authorized staff can create and manage appointments.

22. Payments are recorded after being received at the clinic; no online payment
gateway is included.

23. Payment records must identify the staff user who recorded the payment.

10. Suggested Diagram Deliverables

Diagram Purpose Priority

Use Case Diagram Shows actors and system Essential
functions

ERD	Shows entities, attributes, keys, and database relationships	Essential
Class Diagram	Shows classes, attributes, and object relationships	Essential for design documentation
Activity Diagram	Shows the main case workflow	Essential
Sequence Diagram	Shows interactions between users, UI, backend, and database	Recommended
State Machine Diagram	Shows case statuses and transitions	Recommended
Component Diagram	Shows frontend, API, database, and file storage components	Recommended
Deployment Diagram	Shows runtime environment and deployed components	Optional for MVP

11. Recommended Implementation Order

24. Confirm logical entities and relationships, including Appointment and Payment.

25. Draw the ERD.

26. Review the one-patient/one-tooth Dental Case rule.

27. Draw the Use Case Diagram with Secretary as a distinct actor.

28. Draw the main Activity Diagram.

29. Draw the Class Diagram.

30. Draw Sequence Diagrams for case creation, radiology, design review, appointments, and clinic payments.

31. Draw the State Machine Diagram.

32. Derive the database schema from the approved data model.

33. Review diagrams against the latest SRS before implementation.

End of Revised Analysis and Diagram Design Specification